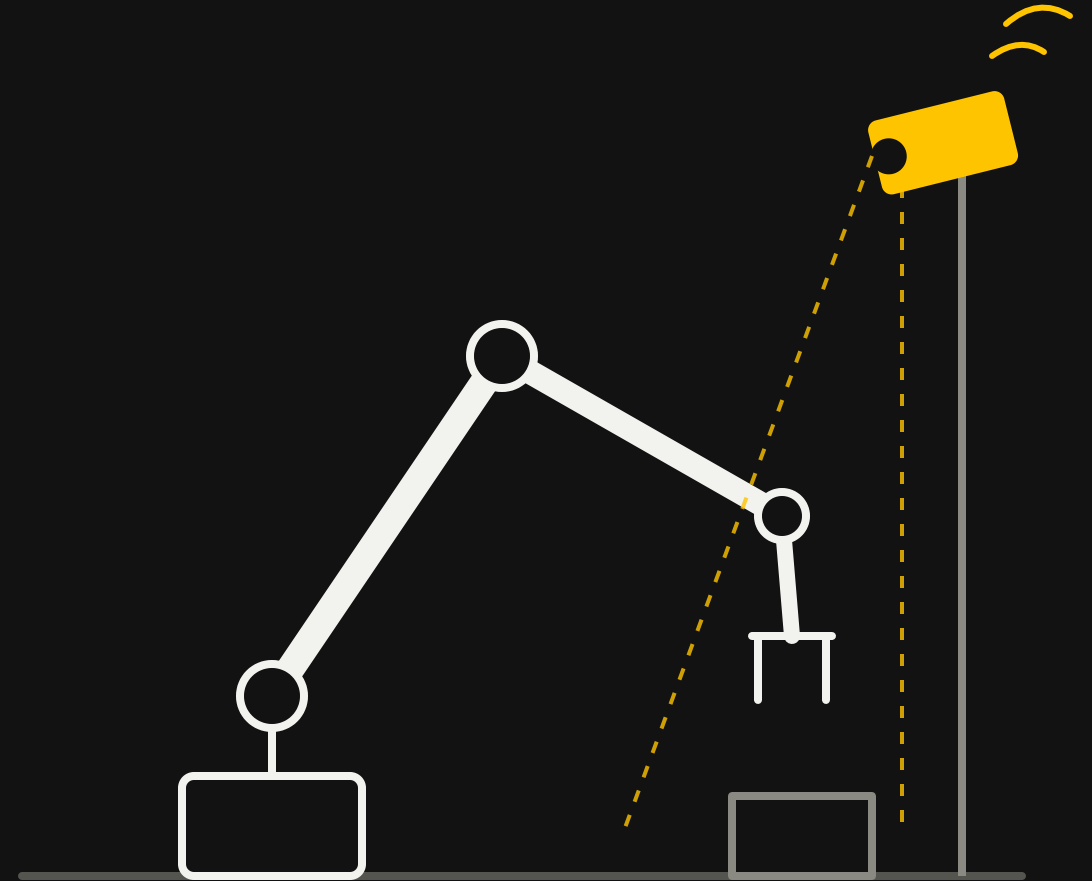


Respite Labs

Software that tells you when a robot arm's camera, gripper or joints have physically moved, by how much, and how to fix it.



A self-check for robot arms.

Respite Labs is software. It runs on the PC that's already next to the robot, so there's nothing new to buy or mount.

Before every shift, and after any bump, the arm runs a six second check motion. Respite Labs reads the arm's own joints and camera while it moves.

If something physical has shifted, it says which part: the camera, the gripper, a tool setting or a joint. It says by how much, in millimeters. Then it writes the correction, checks again, and tells you whether the robot is safe to run.

Between checks it keeps watching and warns when something starts to drift.

WHO IT'S FOR

Anyone running robot arms that have to land in the right place. First: teams training AI on robots, research labs, and factories.

WHAT IT IS NOT

Not a dashboard, not a measuring device, not a new sensor. It answers one question: is the robot's body still where the software thinks it is?

Works in simulation today, with a live demo. Real-arm testing is the next step.

THE PROBLEM

A bumped camera doesn't stop the robot. It makes every grab wrong.

Someone clears a jam and nudges the camera. Two millimeters. No alarm, nothing in the logs. The robot keeps running.

An hour later it's missing parts. Bad parts go down the line, or bad data goes into the training set. The next morning an engineer spends hours working out which of a dozen things changed.

The robot doesn't know it's wrong. Neither does anyone else.

Precision figures: Harazono et al., SLAS Technology, 2024.

0.02 mm

how precisely a robot arm returns to the same spot, every time

2.2 mm

off, and it fails to place a part

The arm isn't wrong. It repeats the wrong spot perfectly.

WHY IT MATTERS

Small physical slips cost real money.

\$2.3M

an hour when a car line stops

\$1.4T

a year in unplanned downtime across the top 500 companies

5M

robot arms working today, and more every year

WHAT KINTRACE CHANGES

A bump is found in 6 seconds, not after a shift of bad parts.

The fix takes minutes, not a technician visit.

Slow drift is caught early, before the first miss.

Bad training data never gets recorded in the first place.

Starting with arms. The same idea applies to any robot with a body.

Sources: Siemens, True Cost of Downtime 2024. IFR World Robotics 2026.

After a bump, today's options take days. Respite Labs takes minutes.

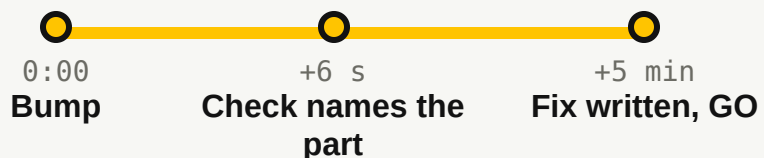
Today

1 to 3 days



With Respite

minutes

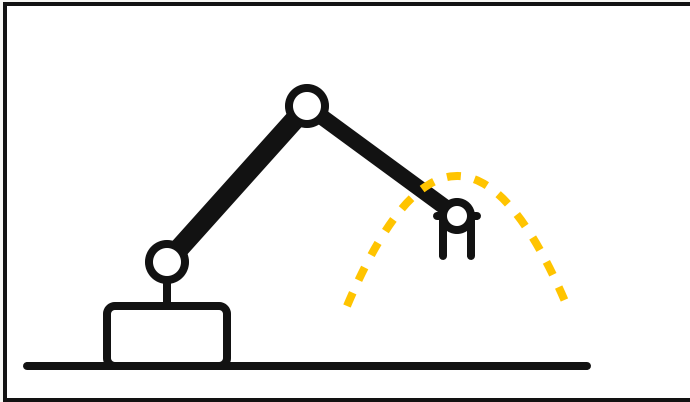


Monitoring software (Ember, Formant, Foxglove) sees error rates go up. It can't say the camera moved.

Measuring hardware (Dynalog, Renishaw) is precise, but a technician has to bring it.

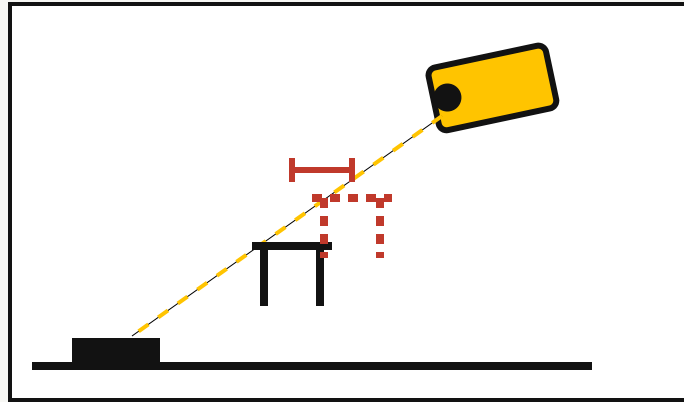
Respite Labs is software only, on the joints and camera the arm already has. Nobody else does the physical check automatically.

Six seconds of motion, one comparison, one answer.



1 · THE CHECK

The arm runs a short, fixed motion, about six seconds, in front of its own camera. No new hardware.



2 · THE COMPARISON

Respite Labs compares where the joints and camera say the gripper is with where it should be. A printed marker on the table is the fixed reference.

```
KINTRACE record, Cell 3  
camera moved 21.2 mm  
gripper bent 24.4 mm  
fix written to controller  
re-check: 6/6 picks ok
```

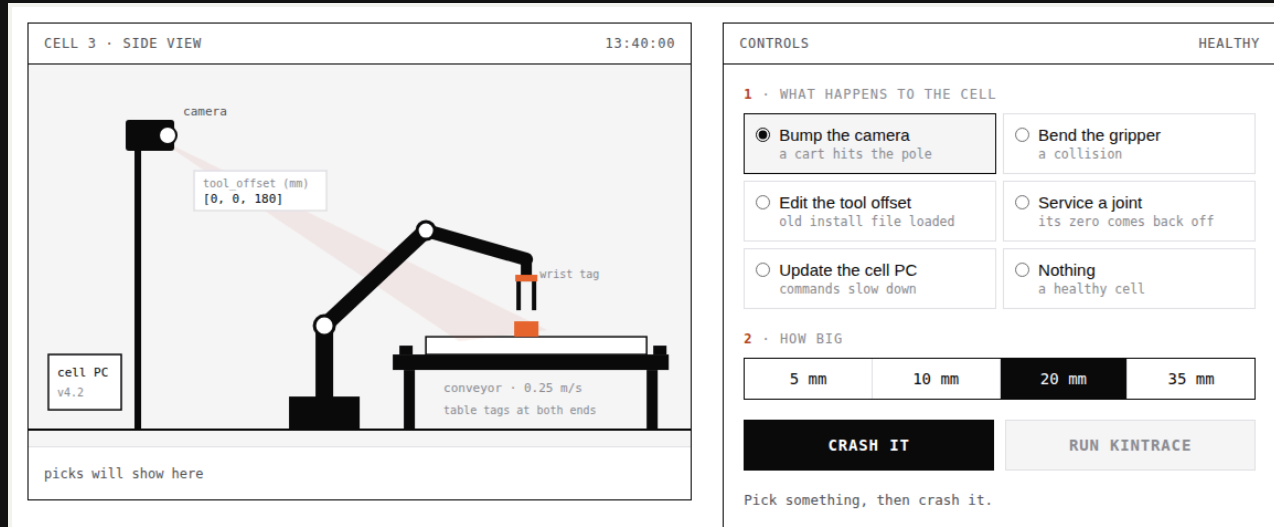
STATUS: GO

3 · THE ANSWER

It names the part that moved and by how much, writes the correction, re-checks, and tells you whether the robot is safe to run.

The record shown is real output from our simulated cell. Runs on the PC already next to the robot.

It's built, it runs, and anyone can try to break it.



Camera moved: 21.2 mm, 0.11 deg
Gripper bent: fingertip moved 24.4 mm
Fix written · re-check ok · test picks 6/6

STATUS: GO

120 / 120

tests where it named the right cause, across six situations

6 seconds

from bump to a written fix and a verdict

Live demo

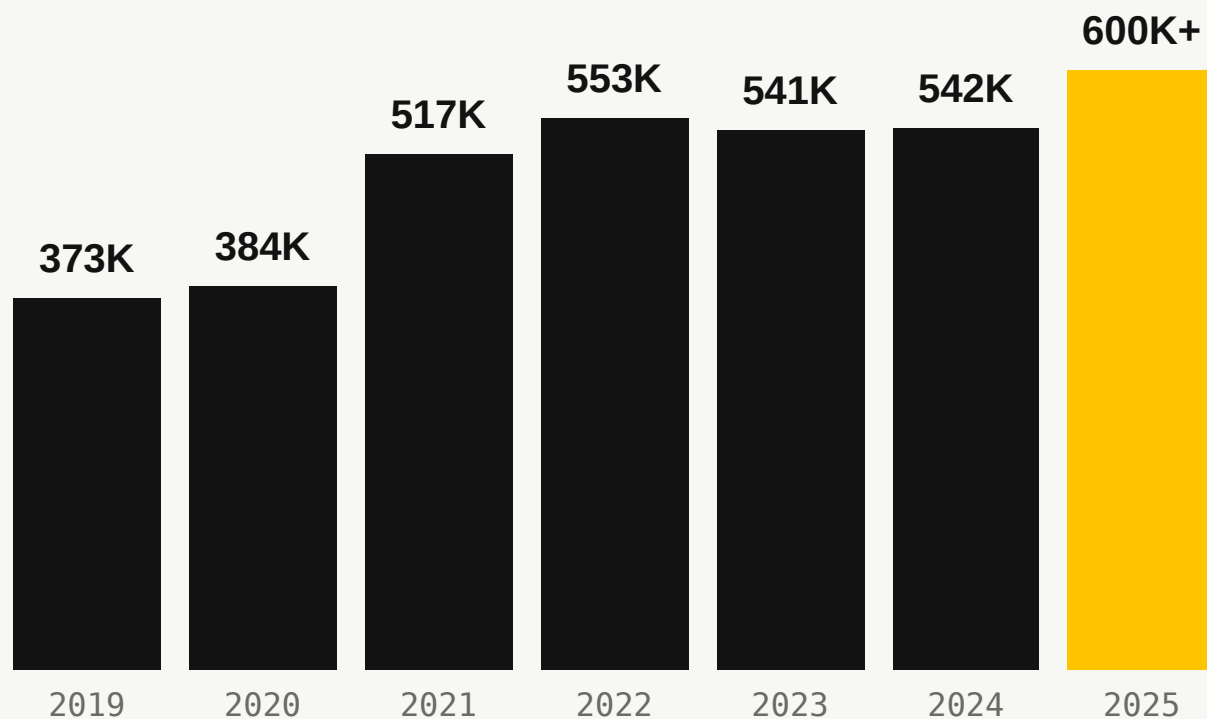
bump the camera or bend the gripper yourself.
Every result is a real run of the code

Open code

about 5,000 lines on GitHub, with tests and a benchmark anyone can re-run

All of this runs on a simulated robot cell we built. Real-arm testing is the next step. github.com/RithikSatarla/kintrace

More arms, less expertise on site, and AI learning from all of it.



New industrial robot installs per year, IFR World Robotics. 2025 rounded.

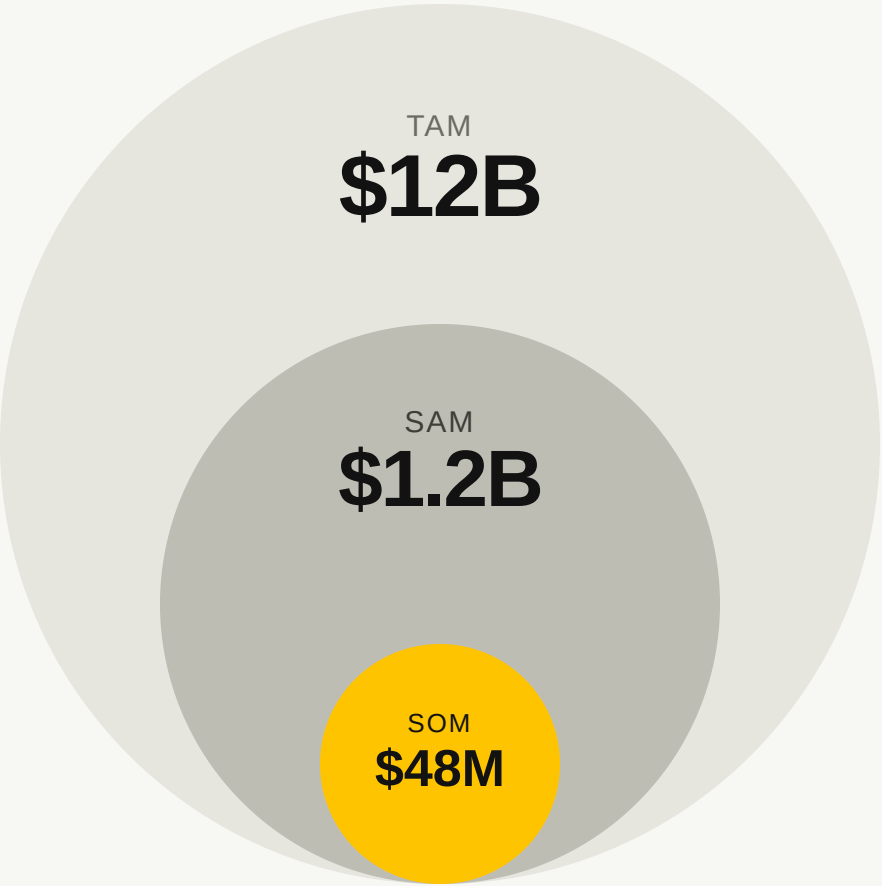
More arms every year, many run by teams with no robotics specialist on site.

AI is being trained on what arms record. A bumped camera puts bad data into every model trained on it.

The check costs nothing extra. The arm already has joints and a camera. The data is already there.

MARKET SIZE

\$12B if every arm ran it. \$48M is our five-year target.



LEVEL	HOW WE GOT IT
TAM total	5M arms × \$200 a month
SAM serviceable	the 10% with cameras today (our estimate)
SOM obtainable	20,000 arms in 5 years

Sources: IFR World Robotics 2026; Market.us robotics services. SAM share and SOM are our own estimates.

One avoided service call pays for a year of it.

Respite Labs, one robot, one year

\$2,400



One technician service call

\$1.2K to \$3K



Yearly upkeep on one \$40K arm

\$2K to \$4.8K



One hour of a stopped car line

\$2.3M



not to scale: this bar would be 1,000 times wider

WHY THEY'D PAY

A robot gets bumped a few times a year. Each one is a service call or an engineer's afternoon.

At \$200 a month, Respite Labs costs less than **one** of those.

For a team training AI, one bad day of data can cost more than a year of the software.

Sources: service call and maintenance ranges from industry maintenance data (PatentPC, 2025); line downtime from Siemens 2024. Price is our planned pricing.

Start free with labs. Sell to the teams whose data it protects.

STEP 1 · NOW

Research labs, free

Labs have arms, break things weekly, and publish. We get real-arm results and the first case studies.

STEP 2 · 6 MONTHS

Teams training AI on robots

They already record everything and pay a lot for clean data. We reach them where they are: the open-source LeRobot community and robot-learning startups.

STEP 3 · 12 MONTHS

Installers and factories

Robot installers (integrators) service hundreds of arms and eat the cost of every site visit. One integrator brings many robots at once.

What gets harder to copy over time: every check on every robot is a labeled record of a real physical fault. Nobody has that data today. It makes the detection better and the early warning earlier.

Two co-founders who built it before asking anyone for money.



Rithik Satarla co-founder

Physical AI research at Carnegie Mellon. Symmetry discovery research at Harvard with Bo Zhao. Built the engine, simulator and demo.



Rajarshi Ghosh co-founder

Harvard CS. Two published AI papers, research with Harvard Medical School, engineering intern at HPE. Built the engine with Rithik and leads labs and early customers.

Working engine

Live demo

120/120 benchmark

Signed records

Desk-arm code

Next: real arms

Respite Labs is a separate project, not affiliated with CMU or Harvard.

WHERE THIS GOES

Real arms this year. Every robot in five.

Year 1

Real arms

Respite Labs running on real robots. 3 pilots with labs and AI teams. First paying customers.

Year 2

First 100 robots

Connectors for the major arm makers. Robot learning teams and labs as paying customers.

Year 3

Fleets

Installers and factories. One customer brings dozens of arms. Early warning across whole fleets.

Year 5

Every robot

The physical health record for robots, beyond arms: mobile robots and humanoids.

WHAT HELPS NOW

Our first engineer, a test arm, and intros to teams running robot arms.

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